

Chunyu Liu

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EDUCATION

IIIS (Yao Class), Tsinghua University

Sep 2024 - Present

B.S. in Computer Science and Technology

Beijing, China

◦ **GPA:** 3.9 / 4.0

◦ **Rank:** 9 / 92

◦ **Selected Courses:**

A+: *Advanced Computer Graphics, Introduction to Large Language Model Applications, Type-safe Modern System Practice, Introduction to Programming in C/C++*

A: *Deep Learning, Machine Learning, Natural Language Processing, Embodied Artificial Intelligence, AI+X Computing Acceleration, Introduction to Computer Systems, CalculusA(1), CalculusA(2)*

PUBLICATIONS

- [1] **VoidPadding: Let [VOID] Handle Padding in Masked Diffusion Language Models so that [EOS] Can Focus on Semantic Termination**
Chunyu Liu, Zhengyang Fan, Kaisen Yang, and Alex Lamb.
Accepted at the COLM 2026 Workshop on Efficient Reasoning (spotlight); under review at NAACL 2027.
- [2] **Frozen in a Frame: The Velocity Blind Spot in JEPA World Models**
Tinghe Zhang, Chunyu Liu, Yu Leon Liu, Zerui Zhao, Jiaheng Chen, Yucheng Xiao, Jiaying Li, Yunlong Wang, and Alex Lamb.
Submitted to ICLR 2027.
- [3] **JustQuant: You Don't Need Smoothing, SVD, or Rotation for 4-Bit Activation Quantization**
Kaicheng Yang*, Kaisen Yang*, Chunyu Liu*, Xianglong Yan, Haotong Qin, Junyi Wu, Tianao Zhang, Xun Zhang, Shaoqiu Zhang, Youbang Sun, and Yulun Zhang.
Submitted to ICLR 2027.
- [4] **Understanding Is Done Early: A Depth Division of Labor in Large Language Models and Its Use for Unbounded-Context Memory**
Hanzuo Liu, Xuan Qi, Chunyu Liu, Haotian Zhong, Yulong Wang, Rayying Key, Alex Lamb, and Mingyu Gao.
Accepted at the COLM 2026 Workshop on Efficient Reasoning; under review at NAACL 2027.
- [5] **A Survey of Efficient Attention Methods: Hardware-Efficient, Sparse, Compact, and Linear Attention**
Jintao Zhang, Rundong Su, Chunyu Liu, et al.
Under review at IEEE TPAMI.

RESEARCH EXPERIENCE

Undergraduate Research Intern, Tsinghua SAIL Group

Jun 2025 – Feb 2026

Supervisors: *Prof. Jianfei Chen* and *Prof. Jun Zhu*

Efficient Attention Methods for LLMs and DiTs

- Investigated linear and sparse attention for large language models and diffusion transformers.
- Co-authored a comprehensive survey [5] and led the linear attention section, analyzing representative architectures and their quality–efficiency trade-offs. 

Undergraduate Research Intern, Tsinghua College AI

Mar 2026 – Present

Supervisor: *Prof. Alex Lamb*

1. VoidPadding for Robust Diffusion Language Models 

- Identified that using [EOS] for both semantic termination and instruction-tuning padding causes [EOS] overflow during large-block decoding in masked diffusion language models.
- Proposed VoidPadding, introducing a dedicated [VOID] token that separates padding from semantic termination and enables adaptive response-canvas expansion.
- Improved the block-size-averaged four-task score on Dream-7B-Instruct by 17.84 points over the base model and 6.95 points over RainbowPadding, while reducing decoding NFE by 55.7%.

2. JustQuant: 4-Bit Activation Quantization for Generative Models

- Ported JustQuant to ELF diffusion language models and built W4A4/W1.58A4 distributed training and multi-seed evaluation; QAD beat direct QAT at all 20 matched-budget 64-step checkpoints under two evaluators.
- Co-developed Theseus QAD, a local-to-global warm-up that progressively expands quantized block groups under

teacher distillation before end-to-end QAT, enabling naive 4-bit activations without smoothing, SVD, or rotations.

3. **CoMem: Depth-Partitioned Memory for Long-Context LLMs**



- Co-developed a depth-partitioned memory architecture that caches mid-layer chunk representations and retrieves a bounded relevant set, keeping model-side read cost independent of stored-context length.
- Achieved 97.05 on RULER and 38.27 on LoCoMo with Qwen3-8B; in a 128K control, reduced peak memory from 89.36 GB to 18.26 GB and accelerated full-write-inclusive prefill by 7.83×.

INDUSTRY EXPERIENCE

Research Intern, LLM Pretraining Algorithms

Jun 2026 – Sep 2026

Ubiquant IQuest Lab

Researching architectural improvements to DeepSeek Sparse Attention for more efficient LLM pretraining.

Technical Staff, Einsia (Remote)

Aug 2026 – Present

- Designed the search functionality for AgentGit; currently help operate and maintain its other platform features.
- Co-developing VideoCut Bench, an ongoing benchmark for teaching agents video-editing capabilities.

PROJECTS

Moe Traits 2 Companion: Multimodal AI Desktop Companion



Computer Vision Course Project (Full Marks)

- Built a PyQt desktop companion integrating screen and camera perception, tool-calling agents, and mem0 for reliable agent memory and context-aware interaction with local content.
- Developed a character-creation workflow with consistent four-emotion sprites, local alpha matting, an interactive workbench, and cross-platform packaging.

Efficient Attention for Diffusion Transformers



Sep 2025 – Feb 2026

Multi-Modal Machine Learning Course Project

- Designed NlogNAttn, an $\mathcal{O}(N \log N)$ attention variant that compresses keys and values while preserving full-resolution queries and 2D RoPE.
- Benchmarked it against full and linear attention on LightningDiT at ImageNet 512×512 , retaining comparable image quality with higher inference efficiency.

PIANO: Character-Aware AI Vocal Cover Pipeline



Deep Learning Course Project (A), Outstanding Demo Project

- Led a 3-member team in building an end-to-end AI vocal-cover system with generated piano accompaniment and character-aware voice conversion.
- Integrated Demucs, DDSP-SVC, Pop2Piano, Beat-This, and FluidSynth into a FastAPI-React pipeline, achieving a 3.5–13.1× speedup.

HACT: Hierarchical ACT Policy for Long-Horizon Embodied Tasks



Embodied AI (A)

- Developed H-ACT with a 4-member team, extending ACT hierarchically for long-horizon language-conditioned manipulation.
- Achieved an 87% simulated success rate, substantially outperforming the 41% ACT baseline.

AWARDS

Silver Medal, 21st Chinese Girls' Mathematical Olympiad (CGMO)

Aug 2022

Freshman Scholarship, Tsinghua University

2024–2025

Artistic Excellence Scholarship, Tsinghua University

2024–2025

SKILLS

Programming: Python, C/C++, Scala, TypeScript, HLSL, Bash, $\text{\LaTeX}$

Tools: PyTorch, Hugging Face Transformers, Linux, Git, Slurm, Docker

Languages: Chinese (native), English (TOEFL 108)